Assignment- Classification

A requirement from the Hospital, Management asked us to create a predictive model which will predict the chronic kidney disease (CKD) based on the several parameters. The Client has provided the dataset of the same.

1.) Identify your problem statement
Using Classification to find kidney disease

2) Tell basic info about the dataset

Input – Age, BP, Protein, Sugar, Blood glucose, Blood urea, Creatinine, HP, Packed cell value, WBC, RBC, Hypertension, Diabetes, Anaemia

3) Mention the pre-processing method if you're doing any (like converting

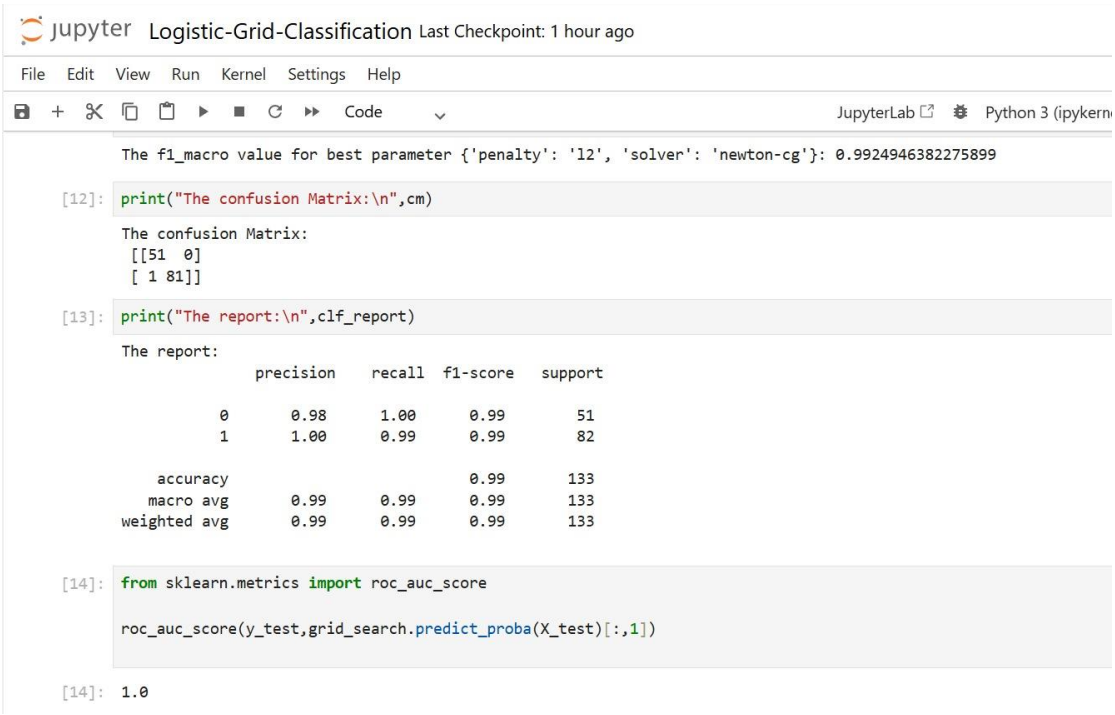
string to number – nominal data)

Use get_dummies for converting string to number and using standard scalar to scale the values

4) Develop a good model with good evaluation metric. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

Logistics ('penalty': 'l2', 'solver': 'newton-cg'), ROC = 1, Accuracy = 0.99

5) All the research values of each algorithm should be documented. (You can make tabulation or screenshot of the results.)



Assignment- Classification

jupyter SVM-Grid-Classification Last Checkpoint: 15 hours ago

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Code

```
cm = confusion_matrix(y_test, grid_predictions)
print(cm)
```

```
[[51  0]
 [ 1 81]]
```

[11]:

```
from sklearn.metrics import classification_report
clf_report = classification_report(y_test, grid_predictions)
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.98      1.00      0.99        51
           1       1.00      0.99      0.99        82

    accuracy                           0.99       133
   macro avg       0.99      0.99      0.99       133
weighted avg       0.99      0.99      0.99       133
```

[12]:

```
from sklearn.metrics import roc_auc_score

roc_auc_score(y_test,grid.predict_proba(X_test)[:,1])
```

[12]:

```
1.0
```

jupyter Random Forest-Grid-Classification Last Checkpoint: 1 hour ago

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Code JupyterLab Python 3 (ipykernel)

[13]:

```
from sklearn.metrics import f1_score
f1_macro=f1_score(y_test,grid_predictions,average='weighted')
print("The f1_macro value for best parameter {}:".format(grid.best_params_),f1_macro)
```

```
The f1_macro value for best parameter {'criterion': 'gini', 'max_features': 'log2', 'n_estimators': 10}: 0.96992481
20300752
```

[14]:

```
print("The confusion Matrix:\n",cm)
```

```
The confusion Matrix:
 [[49  2]
 [ 2 80]]
```

[15]:

```
print("The report:\n",clf_report)
```

```
The report:
              precision    recall  f1-score   support

           0       0.96      0.96      0.96        51
           1       0.98      0.98      0.98        82

    accuracy                           0.97       133
   macro avg       0.97      0.97      0.97       133
weighted avg       0.97      0.97      0.97       133
```

Assignment- Classification

jupyter Naive baye's Last Checkpoint: 14 hours ago

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Code

[7]:
```
from sklearn.naive_bayes import BernoulliNB
classifier = BernoulliNB()
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)
from sklearn.metrics import classification_report
clf_report = classification_report(y_test, y_pred)
print(clf_report)
print(cm)
```

```
              precision    recall  f1-score   support

           0       0.86      1.00      0.93        51
           1       1.00      0.90      0.95        82

    accuracy                           0.94       133
   macro avg       0.93      0.95      0.94       133
weighted avg       0.95      0.94      0.94       133

[[51  0]
 [ 8 74]]
```

jupyter KNN-Grid-Classification Last Checkpoint: 15 hours ago

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Code

```
The confusion Matrix:
 [[51  0]
 [ 4 78]]
```

[12]:
```
print("The report:\n",clf_report)
```

```
The report:
              precision    recall  f1-score   support

           0       0.93      1.00      0.96        51
           1       1.00      0.95      0.97        82

    accuracy                           0.97       133
   macro avg       0.96      0.98      0.97       133
weighted avg       0.97      0.97      0.97       133
```

[13]:
```
from sklearn.metrics import roc_auc_score

roc_auc_score(y_test,grid.predict_proba(X_test)[:,1])
```

[13]:
```
0.9873266379722621
```

Assignment- Classification



6.) Mention your final model, justify why u have chosen the same.
Logistics- Accuracy – 99%